

Evaluation of the Effect of Outreach Activities on Publicizing Radiolarians in Japan Based on the Analysis of Google Trends of “Radiolaria” in 2012–2022

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Radiolarians hold a high academic value (e.g., geological, paleontological, and biological); however, they are not well known to the public worldwide. In conjunction with the InterRad XV in Niigata 2017 (Fifteenth InterRad, international conference of radiolarians), certain outreach activities were conducted to raise the profile of radiolarians. Changes in the Google Trends of “Radiolaria” in Japanese were analyzed to evaluate the publicity effect. The search frequency of the term “放散虫” (Radiolaria in Japanese) increased in the first half of 2017 when the outreach activities were conducted. Image search frequency increased noticeably in September 2019 when a picture book featuring radiolarians was published with related photograph exhibitions. Outreach activities related to InterRad XV achieved the desired effect of raising the radiolarian profile. Furthermore, the release of the picture book and the organization of the photograph exhibitions may have attracted the interest of people in radiolarian images.

Key words: Radiolarian, Outreach activities, Google Trends, Internet, Web search

1. Introduction

Radiolaria is marine holoplankton widespread in both deep and shallow parts of the oceans, right from the equator to the poles (e.g., Lazarus *et al.*, 2021). Polycystine radiolarians have siliceous shells and, therefore, easily form fossils, which have been recorded from the Cambrian to the Quaternary periods (De Wever *et al.*, 2001). Owing to their long fossil records, wide distribution, and distinguished varied shell forms, they are useful index fossils for marine deposits. Radiolarian zones of several Phanerozoic ages have been constructed (e.g., Riedel and Sanfilippo, 1970; Hollis, 1993; O’Dogherty, 1994; Sugiyama, 1997; De Wever *et al.*, 2001; Aitchison *et al.*, 2017; Matsuoka and Ito, 2019). On the Japanese Islands, high-resolution radiolarian zonation has revised the previously held beliefs of geological history (e.g., Ichikawa *et al.*, 1985; Ichikawa, 1990; Isozaki and Maruyama, 1991; Isozaki *et al.*, 2010). This radiolarian-based revision of geologic history is called the “Radiolarian Revolution” (e.g., Ishigaki and Yao, 1982; Nakaseko, 1984; Sato, 1989; Suzuki and Aita, 2011; Matsuoka and Ito, 2017; Ito *et al.*, 2020b). Artists are interested in the unique and varied forms of radiolarians (e.g., Jungck *et al.*, 2019; Ito *et al.*, 2020a).

Regardless of their academic importance, radiolarians are not well known to the public. InterRad is the interna-

tional association of radiolarists. InterRad also means an international meeting of radiolarians held almost every three years. The Fifteenth InterRad (InterRad XV) was organized in Niigata, Japan, in 2017. Outreach activities were conducted in conjunction with the meetings. They were expected to contribute to raising the profile of radiolarians in Japan, leading to an increase in the number of radiolarian researchers, the diversity of radiolarian studies, and the return of research results to society.

Evaluating the publicity effect of such outreach activities can provide important insights for more effective activities in the future. Quantitative evaluation of the publicity effect is difficult; however, evaluation has recently become easier using web search data such as Google Trends. Furthermore, evaluating the effects of outreach activities for radiolarians with low name recognition is easier than for well-known fossils, which are more susceptible to various influences other than outreach activities.

In this article, we first show the several outreach activities conducted to increase awareness among radiolarians. Then, changes in Google Trends for “Radiolaria” are discussed to analyze the publicity effect.

2. Outreach Activities for Popularizing Radiolarians

2.1 Events related to the InterRad XV

The InterRad XV was conducted in Niigata City, Niigata Prefecture, Japan, from October 20 to November 1, 2017 (Matsuoka and organizing committee of the InterRad XV,



Fig. 1. Twitter (current “X”) account of InterRad XV in Niigata 2017 and the flyer calling for participation. (a) Top page of the Twitter account, opened in March 2017 to publicize the information of the meeting. After the meeting, it became an account for disseminating general information on radiolarians. (b) English flyer calling for participation in the conference. A Japanese version was also distributed by the account.

2018; Matsuoka *et al.*, 2019). A total of 187 participants attended the conference, including 46 scientists from 16 countries (Australia, China, France, Germany, Indonesia, Italy, Korea, Mongolia, the Philippines, Russia, Slovenia, Spain, Switzerland, Turkey, the United States, and Japan) attended the conference. In the oral and poster sessions of the meeting, a total of 128 papers were discussed (Ito *et al.*, 2017).

Although InterRad is an international academic meeting for scientists, an appeal was made at the meeting to non-specialists to raise the profile of the radiolarians. A Twitter (current “X”) account was created to publicize the information on radiolarians (Fig. 1a). This account was used to advertise the InterRad XV (Fig. 1b) and announce certain events related to the meeting.

From July 22 to 23, 2017, the 4th Hakubutsu Festival was held at the Science Museum in Tokyo. The Hakubutsu Festival is an annual trade fair for the general public, where Hakubutsu (natural history)-related items are bought and sold. One symbolic object is chosen as the motif of each fair, and radiolarians were selected as the motif for the 4th Hakubutsu Festival. During the festival, A. Matsuoka delivered four lectures to raise the profile of radiolarians and advertise InterRad XV.

To encourage the participation of non-specialists in InterRad XV, two outreach programs were organized at the beginning of the meeting: on October 22, “Let’s make friends with plankton” at TOKIMATE, a satellite (Eki-nan) campus of Niigata University from 14:00 to 15:00 and “What’s

rad???” at the Café Space in JUNKUDO Bookstore Niigata from 15:30 to 17:00 (Fig. 2a, b). In the first event, participants used a microscope to observe plankton with living radiolarians from the Sea of Japan off Sado Island, Niigata Prefecture, assisted by a staff member at the Sado Marine Station of the Faculty of Science, Niigata University. The second event was a scientist–artist collaboration assigned to the 101st performance of Science Café Niigata and was performed by Matsuoka as a scientist and H. Yokoyama as an artist.

2.2 Motif craftworks for arts

Radiolarians are microscale organisms (generally 100–500 μm in shell diameter or width) and are impossible to observe with the naked eye or hold in the hand. Yokoyama created hand-sized craftworks of radiolarians composed of silver. The lineup of the craftworks ranges from the Cambrian to the modern in age (Fig. 3). They were precisely reproduced to retain the maximum original scientific information as possible (Ito *et al.*, 2020a). The silver craftworks are available for sale on RC GEAR website (http://www.rcgear.jp/top_radiolaria.html). The craftwork pieces were distributed as commemorative gifts as part of a welcome pack to the participants of the InterRad XV. In addition to the silver craftworks, other materials, such as beads and cloth, were used for radiolarian craftworks (Ito *et al.*, 2020a).

2.3 Picture book

In 2019, Kanchiku and Matsuoka (2019) published a picture book featuring radiolarians (Fig. 4a). This book

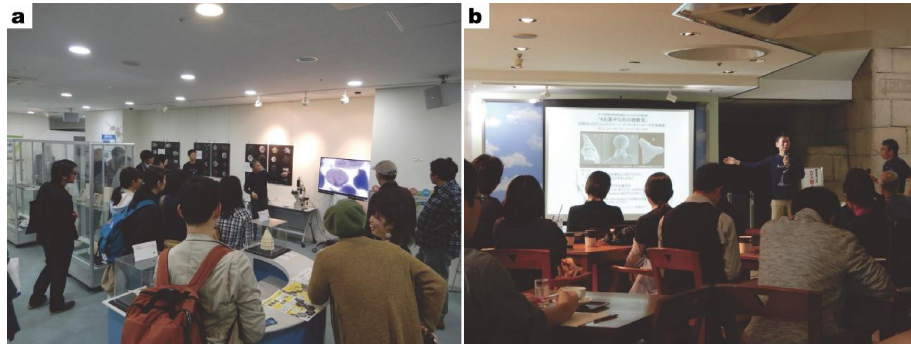


Fig. 2. Lectures held in 2017 related to the InterRad XV in Niigata 2017 (reprints from Matsuoka and organizing committee of the InterRad XV, 2018). (a) Hands-on display “Let’s make friends with plankton” in the Multi-purpose Space of TOKIMATE, a satellite (Eki-nan) campus of Niigata University from 14:00 to 15:00 on October 20, 2017. (b) Talk show “What’s rad??” at Café Space in JUNKUDO Bookstore Niigata from 15:30 to 17:00 on October 20, 2017.

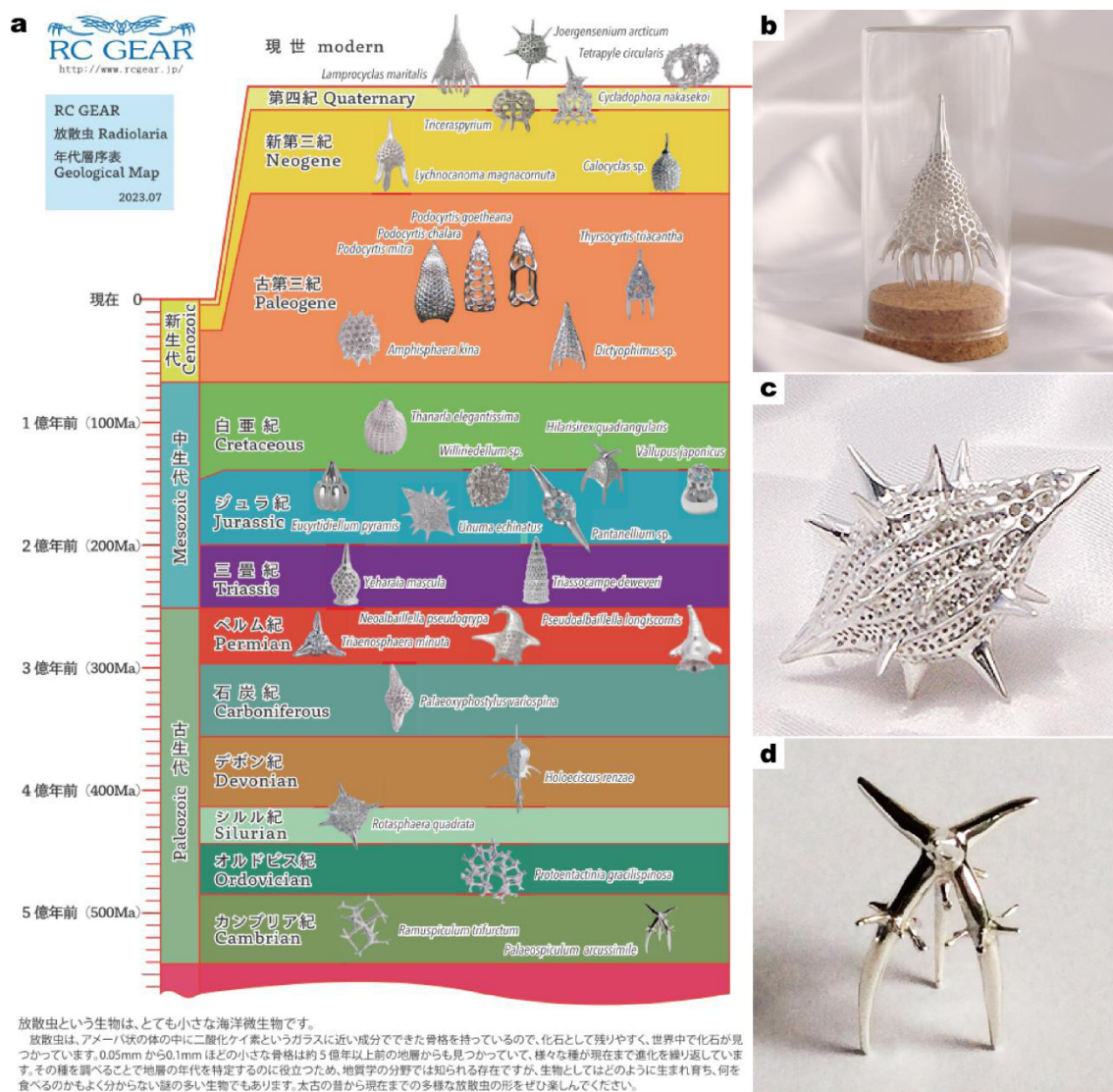


Fig. 3. Silver craftworks depicting radiolarians across several species created by H. Yokoyama through the Phanerozoic era with a geologic time scale. (a) Geologic time scale with radiolarian silver craftworks. (b) *Lamprocyclus maritilis* Haeckel, 1887, exists in modern oceans. (c) *Unuma echinatus* Ichikawa and Yao, 1976, Jurassic species. This species was a symbolic radiolarian at the InterRad XV in Niigata in 2017, and a simplified design based on this species was used as the logo of the meeting. (d) *Palaeosculum arcussimile* Won and Below, 1999, Cambrian species.

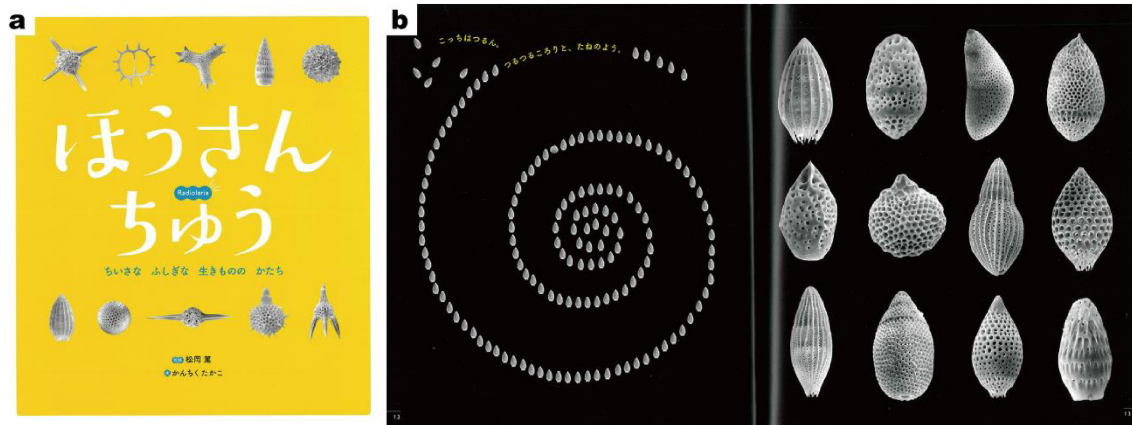


Fig. 4. Picture book featuring radiolarians with sample pages. (a) Cover of the book. (b) Sample pages (pp. 12–13) illustrating the SEM images of several nassellarians.

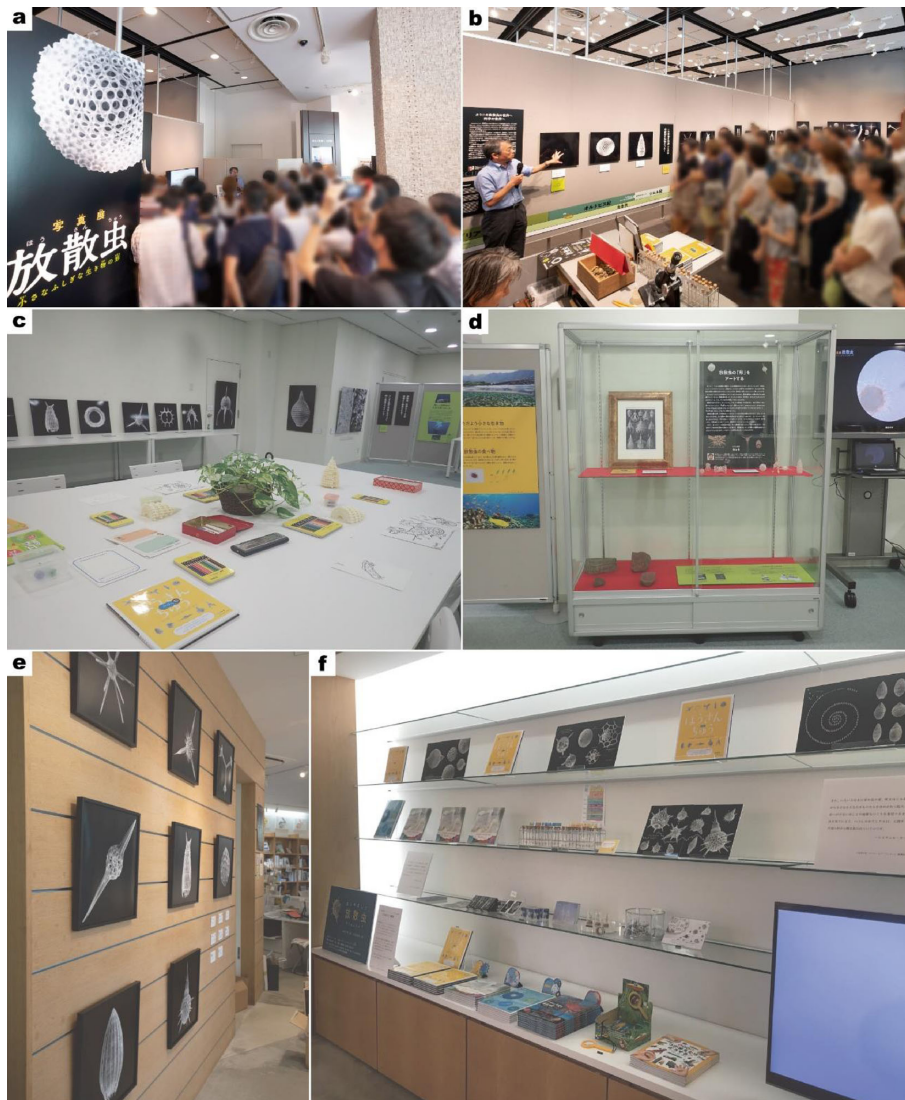


Fig. 5. Photograph exhibitions held in connection with the release of the picture book. (a, b) Lecture event at FUJIFILM SQUARE, Tokyo. (c, d) Photograph exhibition at TOKIMATE, a satellite (Eki-nan) campus of Niigata University. (e, f) Photograph exhibition at Museum shop NOHARA of Vangi Sculpture Garden Museum in Shizuoka.

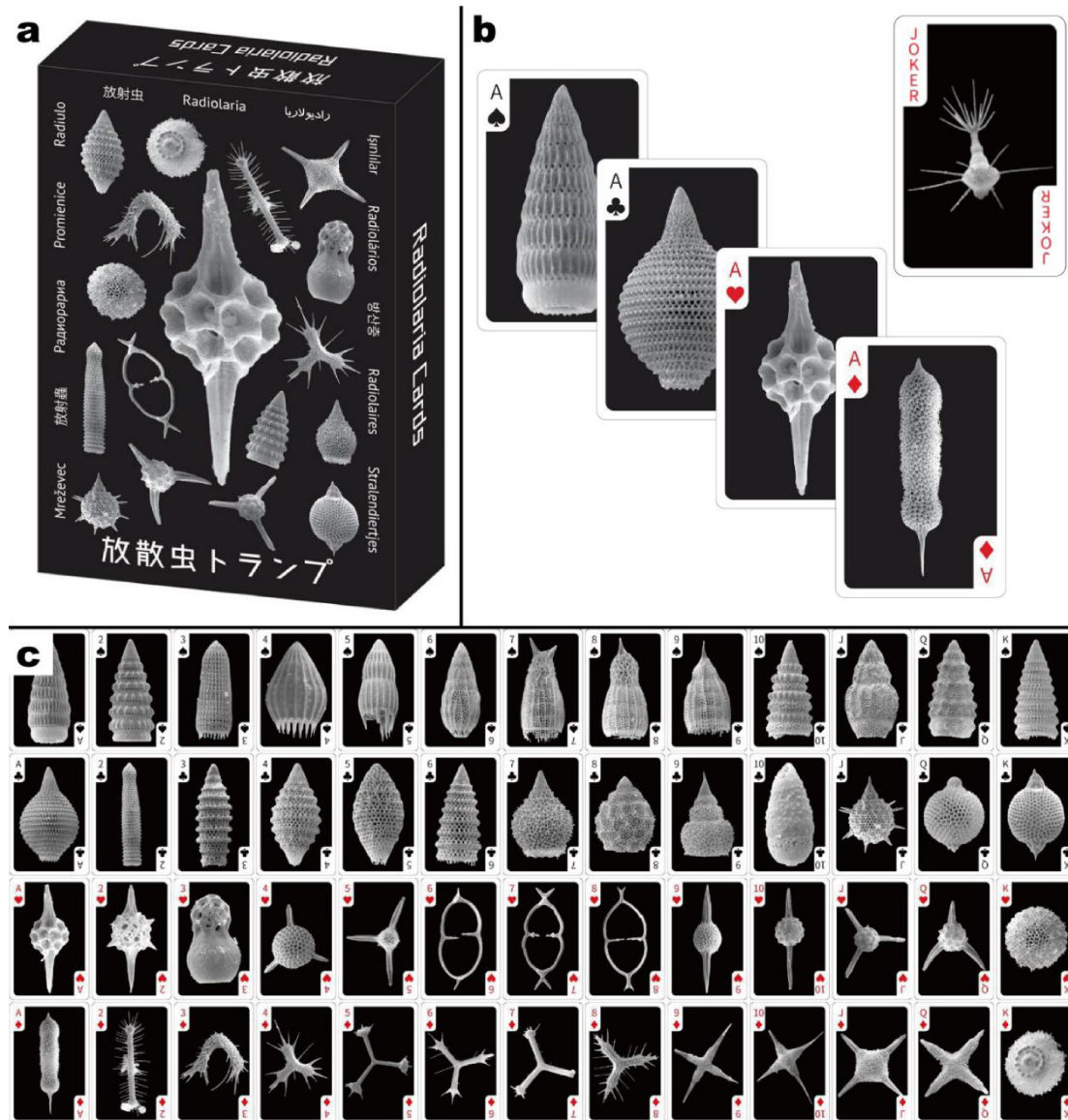


Fig. 6. Playing cards with radiolarian theme. (a) Outer case. (b) Aces of each suit and Joker. (c) All cards except Joker.

presents scanning electron microscopy (SEM) images of the extremely well-preserved earliest Cretaceous radiolarians (Fig. 4b). The radiolarians were obtained from a laminated tuffaceous radiolarian claystone or clayey radiolarite sample collected from the Mariana Trench using a submersible “Shinkai 6500” from the Japan Agency for Marine-Earth Science and Technology (JAMSTEC). The specimens in the picture book were selected mainly from those in Matsuoka (1998). In total, the picture book has many SEM images of different radiolarian species including specimens that were not shown in Matsuoka (1998). An outline of the radiolarians (e.g., body system and feeding behavior) and the sample are described in the book. For the publication of the picture book, Matsuoka provided the SEM images with scientific supervision, T. Kanchiku wrote the text, and T. Kawashima and Kanchiku planned and edited the book.

Furthermore, Kawashima and Kanchiku planned photographic exhibitions concerning the release of the picture

book (Fig. 5). At the exhibitions, the radiolarian SEM images used in the picture book were printed on high-resolution panels and displayed. The first exhibition was held from August 9 to 22, 2019, at FUJIFILM SQUARE, Tokyo. Matsuoka delivered a lecture there (Fig. 5a, b). A total of 28,560 people visited the exhibition during the 14 days. The exhibition traveled to several locations, such as FUJIFILM PHOTO SALON OSAKA (September 13 to 19, 2019), TOKIMATE, the Eki-nan Campus of Niigata University (October 9 to 31, 2019) (Figs. 5c, d), the National Museum of Nature and Science in Tokyo (December 17, 2019 to March 1, 2020), the Museum shop NOHARA of Vangi Sculpture Garden Museum in Nagaizumi, Shizuoka (August 6 to September 29, 2020) (Figs. 5e, f), Toyama Science Museum (August 29 to September 30, 2020), and the Fossa Magna Museum in Itoigawa, Niigata (April 24 to May 30, 2021).

2.4 Merchandise

Several merchandise items related to radiolarians, such

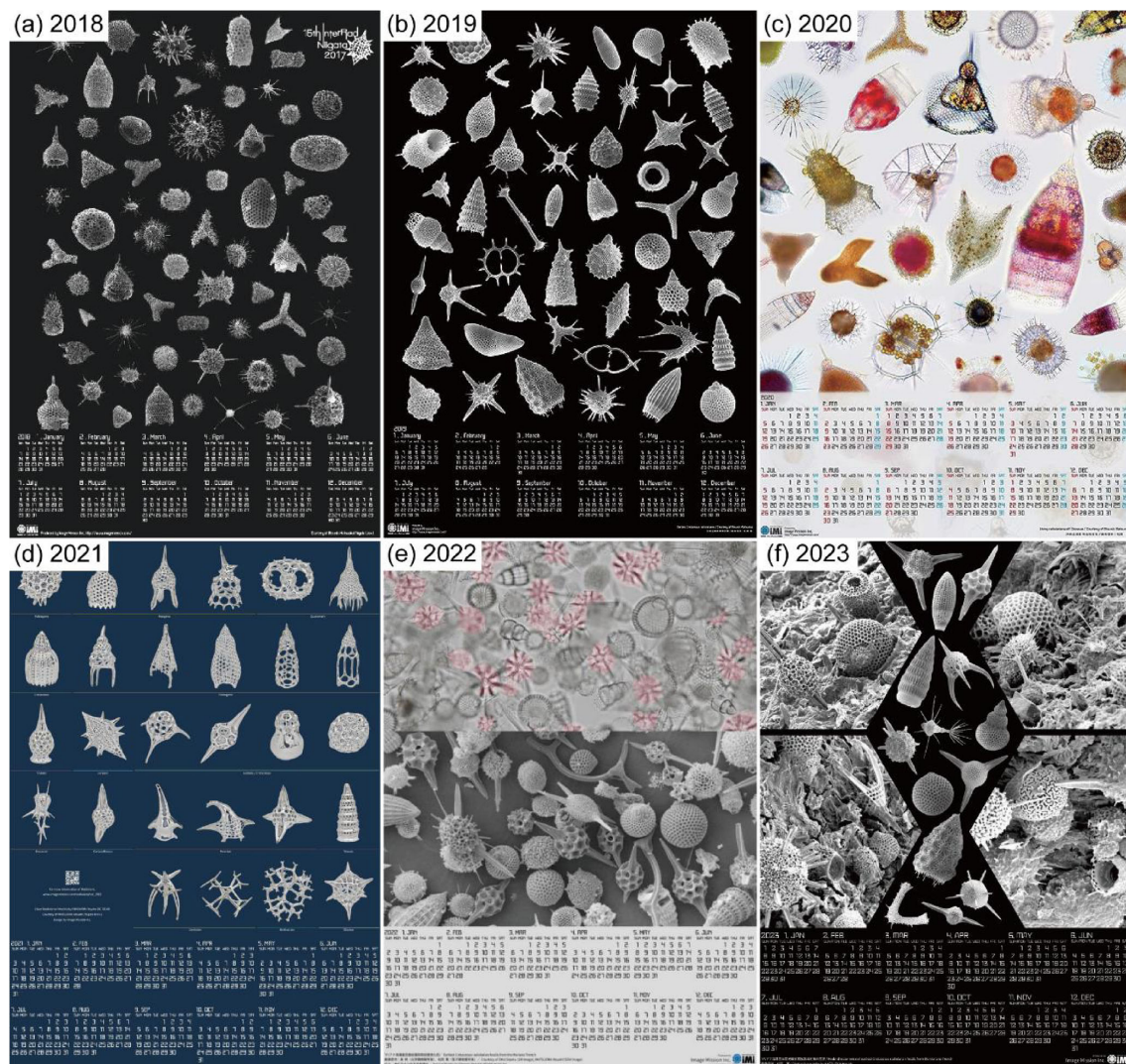


Fig. 7. Radiolarian calendars from 2018 to 2023. (a) SEM images of recent radiolarians from surface waters in the East China Sea. (b) SEM images of the earliest Cretaceous radiolarians from the Mariana Trench. (c) Light microscopy images of living radiolarians in surface waters in the East China Sea. (d) Silver craftworks created by Yokoyama depicting radiolarians from the Cambrian to the Quaternary. (e) Light microscopy and SEM images of the earliest Cretaceous radiolarians from the Mariana Trench. (f) SEM images of specimens on etched surfaces and isolated specimens of the earliest Cretaceous radiolarians from the Mariana Trench.

as playing cards (Fig. 6), calendars (Fig. 7), plastic sleeves, and antiviral mask holders, were made for sale by Image Mission Inc. and they are still being sold today. The playing cards and calendar are shown in this section.

The radiolarian playing cards (Fig. 6a) are available for sale since 2018. The playing card pattern featured SEM images of the earliest Cretaceous radiolarians from the Mariana Trench, which are used in the picture book. The suit of cards reflects a higher classification; that is, specimens depicted on spade and club (black) cards are nassellarian species, whereas those on heart and diamond (red) cards are spumellarian species (Fig. 6b, c) (Matsuoka *et al.*, 2022). The species names on each card are listed on the web page of the Image Mission website (https://www.imagemission.com/radiolaria_d/cards/).

A series of radiolarian calendars have been produced every year since InterRad XV (Fig. 7). They are available for sale via the Internet on the web page of Image Mission Inc. and can be purchased from major museums and cer-

tain bookstores in Japan. The radiolarians featured on the calendar vary annually.

3. Method for Google Trends Analysis

Google Trends is an online tracking system of Internet hit search volumes that has recently merged with its sister project, Google Insights for Search (Google Inc.). A few researchers have used Google Trends to explore the impact of their study objects (*e.g.*, Alicino *et al.*, 2015; Adebayo *et al.*, 2017; Hu *et al.*, 2018), including the COVID-19 pandemic (*e.g.*, Hoerger *et al.*, 2020; Satpathy *et al.*, 2021). In this study, we used Google Trends to explore the effects of outreach activities on radiolarian publicity.

Google Trends provides a relative search volume (RSV), which is the query share, from 0 to 100. An RSV of 100 indicates the maximum search interest during the reference period, whereas 0 indicates an undetectably small value.

Google Trends was mined for 10 years (June 2012 to May 2022), including before and after InterRad XV and re-

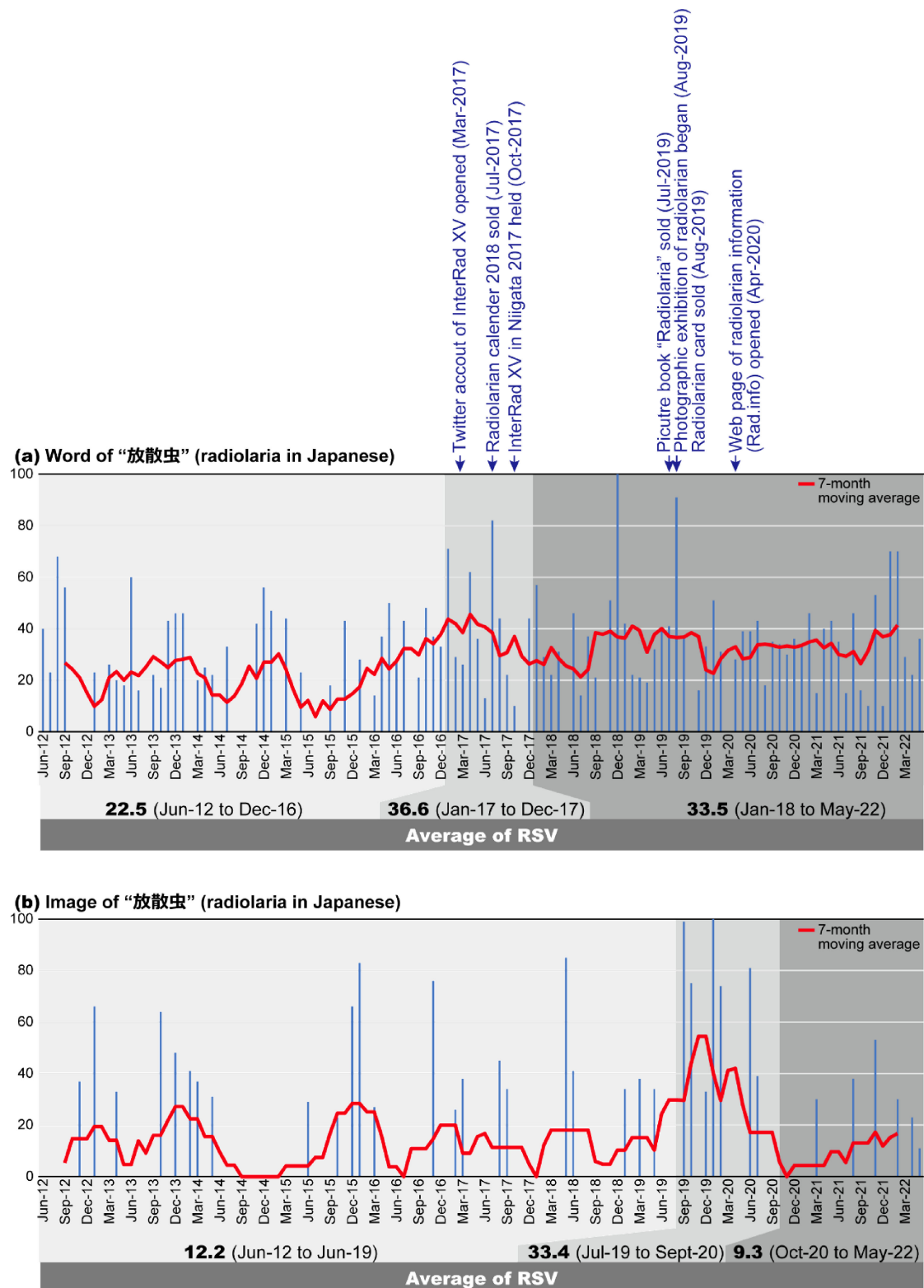


Fig. 8. Relative search volume (RSV) and curve of the 7-month moving average of Google Trends for “Radiolaria” in Japanese from June 2012 to May 2022 (searched in Japan). (a) Search result for the term. (b) Search result for image.

lated outreach activities, to examine their effects. Search words for the term “放散虫” (Radiolaria in Japanese) and its images were developed in Japanese for the web in Japan. Search data were exported from Google Trends in .csv format and later converted into .xlsx format for analysis.

4. Results and Discussion

The search results for the term and images are shown in Fig. 8(a) and Fig. 8(b), respectively. The red lines indicate the 7-month moving average. The average RSV for specific periods is discussed.

In 2017, the number of the term searches increased

[Fig. 8(a)]. The average RSV in 2017 was 36.6. The average RSV up to 2016 was 22.5 and 33.5 after 2018. On average, the RSV increased by *ca.* 1.5 times since 2018 compared with that before 2017 [Fig. 8(a)].

Meanwhile, the number of image searches exhibited no discernible increase around InterRad XV in 2017 [Fig. 8(b)]. Image searches increased in September 2019, when the picture book was published alongside the first photograph exhibition. From July 2019 (when the picture book began to sell) to September 2020 (when the photograph exhibitions took place in Shizuoka and Toyama), the average RSV was 33.4. The average RSV before and after this period was approximately 10 [Fig. 8(b)].

As mentioned in the Introduction, radiolarians are not as well-known publicly as they are academically important. In conjunction with InterRad XV, certain outreach activities were undertaken to raise the profile of radiolarians, and several merchandises related to radiolarians were made available for sale in 2017. Furthermore, releasing the picture book with the photograph exhibition may have aroused interest in radiolarian images. Recently, new approaches to paleontology have been proposed, such as virtual fossils (Rahman *et al.*, 2012) and photogrammetric fossil models (Cunningham, 2021). A diverse and continuous approach is required to publicize the subject and reach the masses. Radiolarian Day (April 30) was established in Japan in 2020, and events for the Radiolarian Day have been held annually since then. On April 30, 2020, a website with general information on radiolarians (Rad.info.: <https://sites.google.com/view/radinfo/>) was released and the website introduces the events.

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